PNAS RESEARCH ARTICLE | SUSTAINABILITY SCIENCE

Molecular-level design of alternative media for energy-saving pilot-scale fibrillation of nanocellulose

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Edited by Alexis T. Bell, University of California, Berkeley, CA; received March 11, 2024; accepted July 23, 2024

The outstanding mechanical properties, light weight, and biodegradability of cellulose nanofibrils (CNFs) make them promising components of renewable and sustainable next-generation reinforced composite biomaterials and bioplastics. Manufacturing CNFs at a pilot scale requires disc-refining fibrillation of dilute cellulose fibers in aqueous pulp suspensions to shear the fibers apart into their nanodimensional forms, which is, however, an energy-intensive process. Here, we used atomistic molecular dynamics (MD) simulation to examine media that might facilitate the reduction of interactions between cellulose fibers, thereby reducing energy consumption in fibrillation. The most suitable medium found by the simulations was an aqueous solution with 0.007:0.012 wt.% NaOH:urea, and indeed this was found in pilot-scale experiments to reduce the fibrillation energy by ~21% on average relative to water alone. The NaOH:urea-mediated CNFs have similar crystallinity, morphology, and mechanical strength to those formed in water. The NaOH and urea act synergistically on CNFs to aid fibrillation but at different length scales. NaOH deprotonates hydroxyl groups leading to mesoscale electrostatic repulsion between fibrils, whereas urea forms hydrogen bonds with protonated hydroxyl groups thus disrupting interfibril hydrogen bonds. This suggests a general mechanism in which an aqueous medium that contains a strong base and a small organic molecule acting as a hydrogen-bond acceptor and/or donor may be effectively employed in materials processes where dispersion of deprotonable polymers is required. The study demonstrates how atomic-detail computer simulation can be integrated with pilot-scale experiments in the rational design of materials processes for the circular bioeconomy.

biomass | nanocellulose | fibrillation

Significance

Nanocellulose from biomass has outstanding mechanical properties and is promising for manufacturing sustainable composite biomaterials and bioplastics for the circular bioeconomy. However, obtaining nanocellulose at pilot scale requires energy-intensive fibrillation to shear cellulose fibers apart into nano-dimensional forms in water. We used atomistic simulation to predict whether alternative media can reduce interactions between cellulose fibers, thereby potentially reducing energy consumption in fibrillation. The most suitable medium found in the simulations, aqueous NaOH:urea (0.007:0.012 wt.%), is found experimentally to reduce the fibrillation energy by ~21% on average relative to water alone. This study also suggests that water containing a strong base and a molecule with hydrogen-bond acceptor/donor capability may be applied to materials processes where dispersion of deprotonable polymers is needed.

Cellulose, a major plant cell wall biomass component, is the most abundant biopolymer on Earth (1). In plants, cellulose nanofibrils (CNFs) are found with individual widths ~10 to 100 nm (2). Each CNF contains multiple cellulose elementary fibrils (CEFs), which are semicrystalline and have been widely accepted to have 18 glucan chains each (*SI Appendix*, Fig. S1) (3–5). CNFs have a broad range of applications including in reinforced composites, bioplastics, packaging, construction, electronics, and sensors (6–9). The biodegradability, abundant availability, amenability to surface modification to tailor properties (10), light weight, and outstanding mechanical properties (11) of CNFs make them very promising components of next-generation composite materials for the circular bioeconomy.

The use of CNFs in additive manufacturing (AM) (3D printing) is of particular interest. AM has recently been at the forefront of various applications such as in biomedical, construction, and aerospace industries (12). AM of fiber-reinforced composites, in which fibers are dispersed in a polymer matrix, has been emerging in recent years as these composites have high strength-to-weight ratios and are lightweight (13). In large-scale AM, carbon fiber–reinforced acrylonitrile butadiene styrene (CF-ABS) is widely used but is, however, not renewable (14). Therefore, there has been increasing demand for replacing CF-ABS with lignocellulosic fiber (15)-reinforced biopolymers such as polylactic acid (PLA) (16) [i.e., materials made from plant biomass, a renewable and sustainable resource (17)]. In one example, AM of CNF-reinforced PLA composites has been applied to build a 600-square-foot house (18).

Obtaining CNFs requires separating them from plant materials using mechanical fibrillation techniques such as disc refining, grinding, and extrusion (19), which results in dilute aqueous suspensions with 2 to 5 wt.% CNFs (10). Among these, disc refining, a pilot-scale process, is the most energy-efficient method (20). In a disc refiner, cellulose fibers in pulp, hundreds of μm in length, are sheared into CNFs at ambient temperature

The authors declare no competing interest.

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This article contains supporting information online at https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2405107121/-/DCSupplemental.

Published September 5, 2024.

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using water as a medium (21). However, disc refining still consumes thousands of kW h energy per metric ton of CNFs (21), and the overall energy consumption of manufacturing biodegradable CNF products is still approximately twice that of competing fossil-based polyethylene terephthalate (PET) products (22).

To reduce energy consumption in fibrillation, chemical pretreatments of pulp have been studied and reviewed extensively (23, 24) and are promising at a bench laboratory scale but have not yet been successfully implemented at a pilot plant scale due to their own safety, toxicity, and optimal performance issues at elevated temperatures (23). Therefore, it is important to rationally search for medium alternatives to water that can be suitably incorporated into pretreating pulp with a disc refiner and facilitate fibrillation. Water-based media that include urea or glycerol have been proved to dissolve cellulose well at ambient temperature (25, 26). In addition, γ-valerolactone (GVL) also dissolves cellulose (27), and its aqueous solution serves as a fractionation medium to produce high-purity cellulose from birch wood (28). These three different molecules may act as hydrogen-bond acceptors to disrupt the association of hydroxyl groups between cellulose fibrils (26). This causes fiber swelling in pulp pretreatment and subsequently helps a disc refiner shear cellulose into its nanodimension using less energy (23). In contrast, cellulose is not soluble in dimethyl sulfoxide (DMSO) at 25 °C (29).

In this work, we examined eight media [two with urea (25), three with glycerol (26), two with GVL (27, 28), and DMSO (29). *SI Appendix*, Table S1]. We performed atomistic MD simulations of cellulose in each of the eight media and in water at 25 °C. Five of the media reduced number of inter-CEF atomic contacts in the simulations as compared with water only. Of these, aqueous NaOH:urea (0.007:0.012 wt.%) was found the most suitable for employment in pretreatment of pulp and disc-refining fibrillation. The experimental results show that using dilute NaOH:urea in water produces CNFs with the same sizes and material properties as water alone and saves ~21% fibrillation energy on average. Our study illustrates the usefulness of approaches based on atomistic simulation to save energy in biomaterials processing.

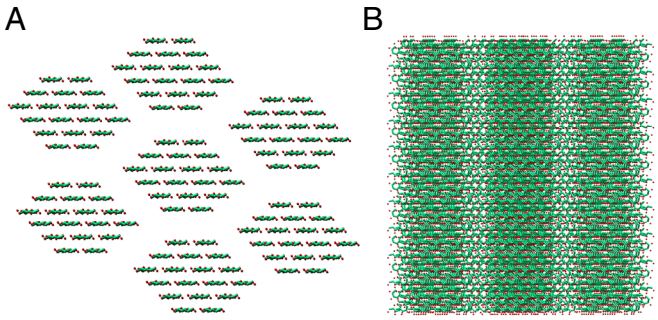
Results

All-Atom MD Simulations. We performed MD simulations of seven bundled CEFs with a 2-3-2 arrangement (30, 31) and degree of polymerization (DP) of 20 (32) (Fig. 1 *A* and *B*) in each of the eight media and in water as a reference at 25 °C. A model of seven CEFs rather than multiple CNFs was implemented for computational reasons since inter-CEF interactions can be extrapolated to inter-CNF interactions, the decrease of which results in fiber swelling in pretreatment of pulp. This, in principle, may lead to lower energy consumption in subsequent disc-refining fibrillation.

The capability of any given medium to disperse the seven CEFs was estimated from the MD simulations as ΔC, which is the difference in the number of inter-CEF atomic contacts in between an examined medium (C_e) and water (C_w) as follows:

$$\Delta C = \frac{C_e - C_w}{C_w}, \qquad [1]$$

where a contact is defined as established when two atoms from adjacent CEFs are within 3.6 Å, approximately the first hydration shell of a CEF (*SI Appendix*, Fig. S2). The atoms calculated excluded the 3 glucose monomers on each end of the glucan chains so as to eliminate end effects and thus approximate the experimental



C

Ranking	Medium	ΔC (%)	ΔC+ (%)	ΔC- (%)
1	Glycerol:water (1:2 vol.)	-3.19	-2.58	-3.93
2	Aqueous NaOH:urea (7:12 wt.%)	-2.92	-2.62	-3.28
3	γ-valerolactone:water (1:1 vol.)	-2.12	-1.29	-3.13
4	Aqueous LiOH:urea (4.2:12 wt.%)	-1.44	-2.11	-0.63
5	Glycerol:water (2:1 vol.)	-0.73	-0.12	-1.48
6	*Water*	*0.00*	*0.00*	*0.00*
7	Glycerol:water (1:1 vol.)	1.00	0.88	1.14
8	γ-valerolactone (GVL)	2.12	1.33	3.08
9	Dimethyl sulfoxide (DMSO)	4.02	4.12	3.89

Fig. 1. Atomistic model of seven CEFs with DP of 20 in cross-sectional (*A*) and top–down (*B*) views. Carbon atoms are green, oxygen atoms are red, and hydrogen atoms and medium species are not shown for simplicity. (*C*) Ranking of the eight examined media by reduction of inter-CEF contacts as compared with water, italicized as a reference, from the MD simulations [ΔC, Eq. [**1**]]. $\Delta C+$ and $\Delta C-$ stand for upper and lower limits of contacts, respectively, considering standard errors among three independent runs. Vol. and wt. stand for volume and weight, respectively. Created with BioRender.com.

conditions, in which CNFs are several μm in length (33). The number of contacts was taken as the average over 12 adjacent CEF pairs (Fig. 1*A*) and three independent runs of the last 100-ns data (*SI Appendix*, Fig. S3).

Fig. 1*C* shows the eight examined media ranked according to ΔC in ascending order (the more negative the ΔC, the better the medium). Taking the statistical uncertainty into account, the top 5 media are predicted to reduce inter-CEF contacts as compared with water. The negative results for GVL and glycerol:water (1:1 vol.) are reasonable, since cellulose is soluble in GVL at 200 °C (27) rather than at room temperature and DMSO does not dissolve cellulose at 25 °C (29) and correspondingly does not reduce inter-CEF contacts (ΔC = 4.02%).

Practical considerations restrict which of the top-ranking media can used in pulp pretreatment and fibrillation. Glycerol:water (1:2 vol.) was excluded since it is ~3 times more viscous than water (34, 35), which increases energy consumption in fibrillation. Also, γ -valerolactone:water (1:1 vol.) is incompatible with the polymeric seals of disc-refining equipment and needs to be removed after fibrillation.

Of the top 3 media, aqueous NaOH:urea (7:12 wt.%) was found suitable for pulp pretreatment and fibrillation but needed to be diluted to aqueous NaOH:urea (0.007:0.012 wt.%) so that its pH was below 9 to avoid corrosion in the pilot-scale disc refiner. Further, use of this diluted NaOH:urea in water allows CNF products to be applied in reinforced composites directly without washing.

Pilot-Scale Fibrillation of Cellulose Fibers in Pretreated Pulp. Pulp was soaked in aqueous NaOH:urea (0.007:0.012 wt.%), or water as a reference, for 12 h as pretreatment. Subsequently, disc-refining fibrillation was performed on cellulose fibers in the pretreated pulp to produce dilute CNFs in suspensions.

A plot of the CNF fines level, which is the wt.% of fibers with length less than 200 μm (21), vs. the fibrillation energy level (Fig. 2*A*) shows that less energy is consumed using the dilute

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NaOH:urea in water medium than in water alone to achieve any given fines level. In other words, given the same energy input, CNFs with higher fines level (i.e., smaller size) are obtained with the NaOH:urea than water only.

Quadratic polynomials fitted to the data in Fig. 2*A* yield:

$$F_A = 0.13E_A^2 + 7.94E_A + 14.15, \quad [2]$$

$$F_W = 0.46E_W^2 + 4.04E_W + 14.21, \quad [3]$$

where F_A and E_A are the CNF fines and fibrillation energy levels, respectively, in aqueous NaOH:urea (0.007:0.012 wt.%), and F_W and E_W are the corresponding quantities in water. R^2 values for Eqs. [**2**] and [**3**] are 0.9899 and 0.9943, respectively. According to these equations, using the dilute NaOH:urea in experiment, ~21% fibrillation energy is saved on average over fines levels of 17 to 100% as compared with water only.

Zeta Potential Measurements of Fibrillated CNFs in Suspensions. To understand the reduced energy consumption using the dilute NaOH:urea in water as a fibrillation medium, a number of experiments and calculations were performed. In one such experiment, the Zeta potential, which is a measure of the electrostatic repulsion between adjacent and similarly charged particles in a dispersion (36), was measured. It was found that the Zeta potential of fibrillated CNFs in aqueous NaOH:urea (0.007:0.012 wt.%) is −44.3 ± 0.2 mV, significantly lower than in water (−23.7 ± 1.5 mV), indicating that CNFs in the NaOH:urea are more negatively charged and thus more repulsive than those in neutral water only. The extra negative charge may result from deprotonation of hydroxyl groups in cellulose by the alkaline NaOH environment (37).

Crystallinity, Morphology, and Mechanical Strength of CNF Products. We next examined whether aqueous NaOH:urea (0.007:0.012 wt.%) produces CNFs with properties similar to or better than those produced in water.

In both media, ^{13}C nuclear magnetic resonance (NMR) spectra of fibrillated CNFs (Fig. 2*B*) exhibit peaks of native cellulose C1 at 104 ppm, C4 (crystalline) at 88 ppm, C4 (amorphous) at 82 ppm, C2, C3, C5 at 74 to 71 ppm, C6 (crystalline) at 64 ppm, and C6 (amorphous) at 61 ppm (38). There is no peak of urea at 165 ppm (39) since its concentration is very low (0.012 wt.%). The cellulose crystallinity index (*CI*) of the CNFs was also calculated according to

$$CI = \frac{A_{86-92ppm}}{A_{86-92ppm} + A_{79-86ppm}}, \quad [4]$$

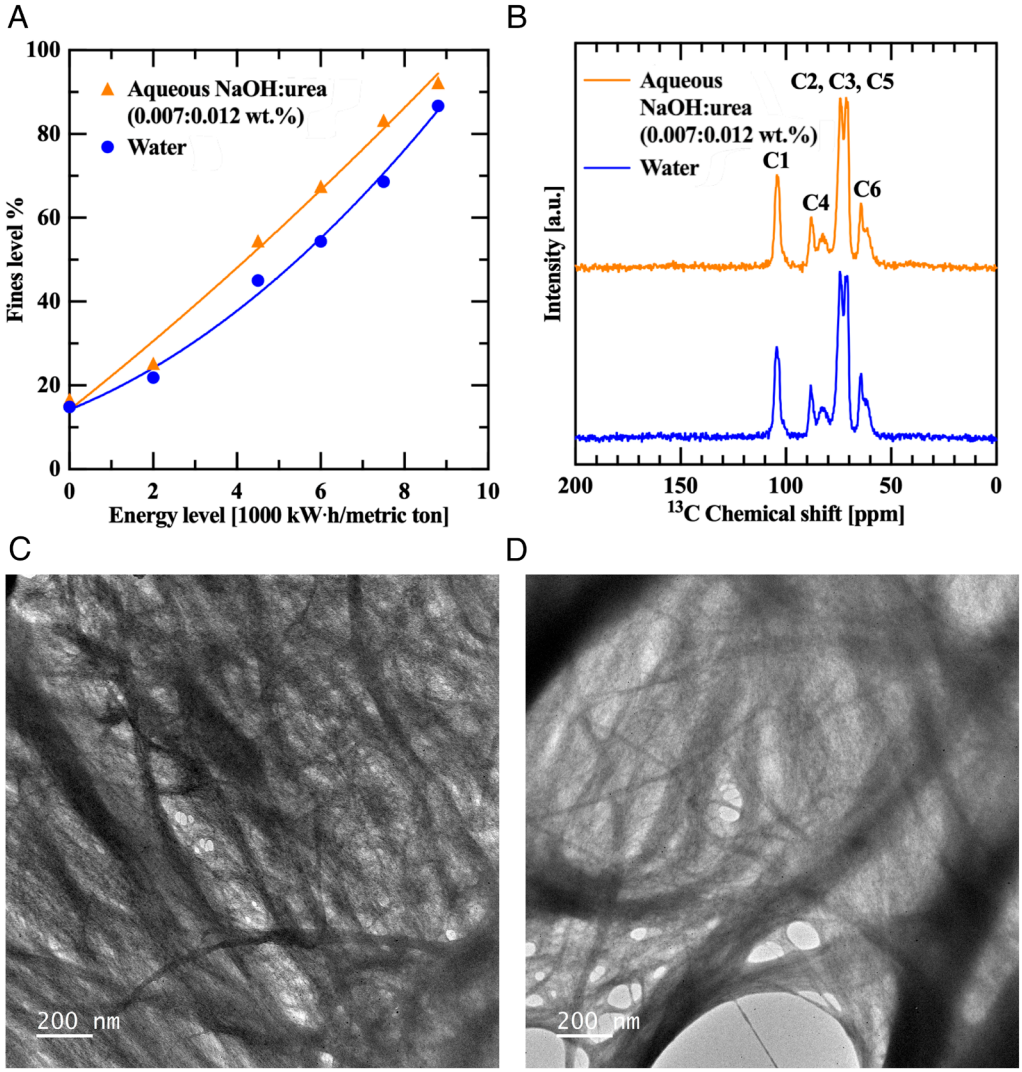

Fig. 2. (*A*) Fines level of produced CNFs vs. fibrillation energy level for pretreated pulp in aqueous NaOH:urea (0.007:0.012 wt.%) and sole water. The formulas of two fitted curves are shown in Eqs. [**2**] and [**3**]. (*B*) ^{13}C NMR spectra of CNFs fibrillated in aqueous NaOH:urea (0.007:0.012 wt.%) and sole water. a.u.: arbitrary unit. TEM images of CNFs fibrillated in aqueous NaOH:urea (0.007:0.012 wt.%) (*C*) and sole water (*D*). Created with BioRender.com.

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where $A_{86-92ppm}$ and $A_{79-86ppm}$ are deconvoluted and integrated areas of the C4 (crystalline) and C4 (amorphous) peaks, respectively (40). The *CI* of CNFs produced in the NaOH:urea (50.4%) is not significantly different from that in water (49.3%).

Transmission electron microscopy (TEM) images (Fig. 2 *C* and *D*) show that CNFs obtained in the NaOH:urea and in water both have similar morphologies and dimensions with widths mostly below 100 nm and lengths of several μm. Further, the tensile index of hand sheets with 95 wt.% hardwood and 5 wt.% CNFs fibrillated in the two media was tested. The tensile index for the hand sheets with the NaOH:urea-mediated CNFs (52.7 N m/g, stretch: 3.21%) was also found to be comparable to that with water-mediated CNFs (53.2 N m/g, stretch: 3.16%).

The experimental results above show that dilute NaOH:urea in an aqueous solution serves as an alternative medium in pilot-scale fibrillation of plant nanocellulose and saves a significant amount of energy in the production of CNFs, which have similar crystallinity, morphology, and mechanical strength to those conventionally mediated by water alone.

Discussion

To understand the role of molecular species in aqueous NaOH:urea in facilitating CEF separation, which corresponds to cellulose fiber swelling in pulp pretreatment for energy-saving fibrillation, further analyses of the MD simulations were performed.

The proximal radial distribution function (RDF) $g(r)$, a measure of the local probability density of atoms (17), between surface of the seven CEFs and the molecular species (Fig. 3*A*) shows that water is the dominant species on the cellulose surface. However, urea also interacts with the cellulose surface via two solvation shells, peaked at 3.6 and 4.8 Å, reducing the local water density without altering its radial solvation shell structure (39). We also find that urea surrounds the seven CEFs homogeneously (Fig. 3*B*), promoting steric hindrance between CNFs and thus cellulose fiber swelling. As expected, the proximal RDF between surface of the central CEF only and urea and water (Fig. 3*C*) is similar to that of the seven CEFs but with the difference of lower peak intensities for all species and no significant second solvation shell for urea, as the inter-CEF space is relatively limited. Nevertheless, urea still binds to the central CEF to some extent (Fig. 3*D*). No significant

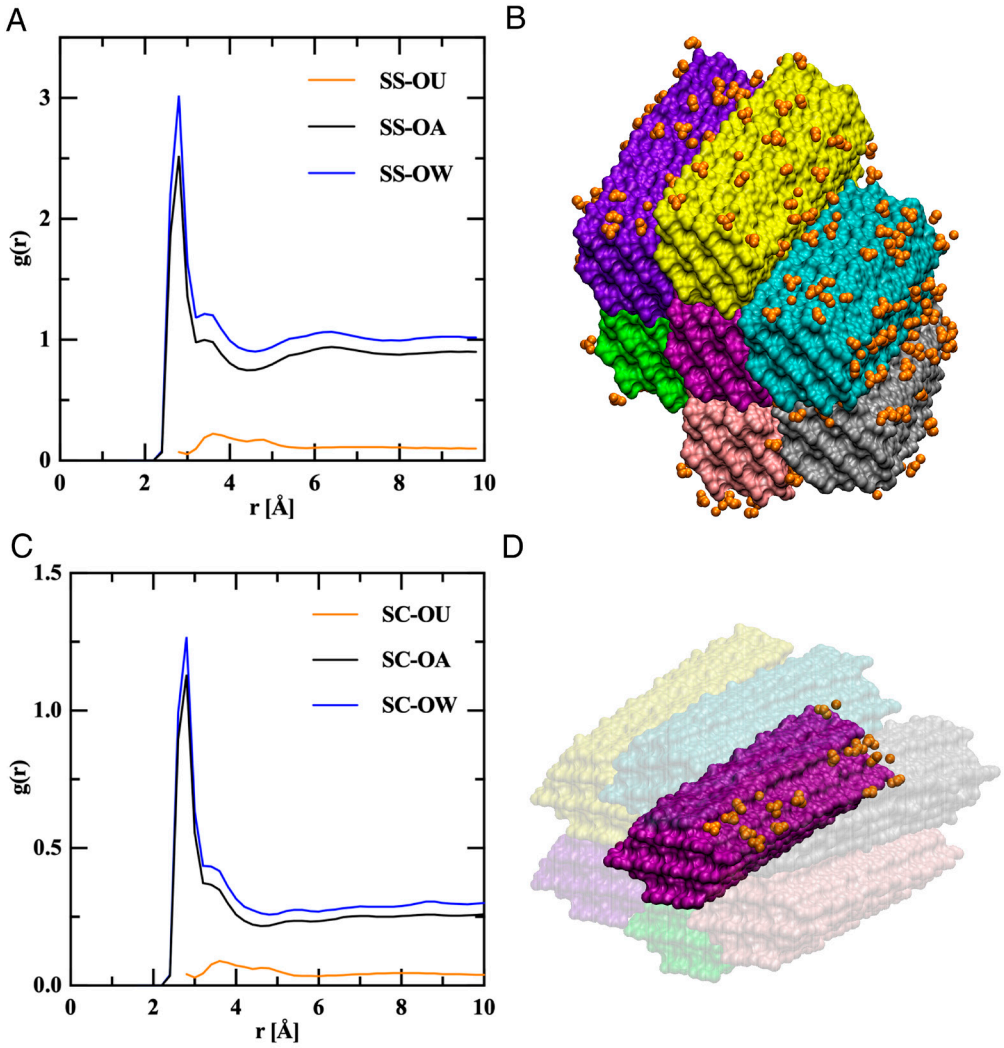

Fig. 3. (*A*) Proximal RDF $g(r)$ between nonhydrogen surface atoms of the seven CEFs (SS) and the oxygen atoms in urea (OU), oxygen in water in aqueous NaOH:urea (7:12 wt.%) (OA), and oxygen in sole water (OW). The values are averaged over three independent MD runs. (*B*) Urea molecules (orange) within 4 Å of the cellulose (seven distinct colors each for one CEF). The 3 glucose monomers on each end of the glucan chains are not shown for simplicity and consistency with analyses. (*C*) Similar to (*A*) except that the reference is the central CEF (SC) only. $g(r)$ for water does not converge to 1 at long distances r since the central CEF is surrounded by six CEFs where local water density is reduced. (*D*) Similar to (*B*) except that only urea molecules within 4 Å of the central CEF are shown and the surrounding six CEFs are transparent. Created with BioRender.com.

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RDF peak is observed between the surface of the seven CEFs and Na^+ or OH^- (*SI Appendix*, Fig. S4), indicating a preference for the bulk for these ionic species. Any potential association between Na^+ and deprotonated hydroxyl groups in the presence of alkaline NaOH (37) cannot be observed in the current simulations since the seven-CEF model is electrostatically neutral and fully protonated. At the pH of the experimental work (below 9), given the pKa of cellulose 13.3 (41), and the fact that only one hydroxyl group per glucose monomer exposed to the external solution is deprotonable (42), then for a model of the size of our MD system, all the hydroxyl groups in our model should still be protonated according to the Henderson–Hasselbalch equation. For a larger CNF, ~20 nm in diameter and ~2 μm in length (i.e., 91 bundled CEFs with a 6-7-8-9-10-11-10-9-8-7-6 arrangement and DP of 4,000), in aqueous NaOH:urea (0.007:0.012 wt.%), there would be at most ~200 deprotonated hydroxyl groups (i.e., ~100 times more than in neutral water). Therefore, depending on the dielectric screening present, there may be a mesoscopic electrostatic repulsion consistent with the Zeta potential results in addition to the atomic-scale observations from the MD.

Urea is found to form hydrogen bonds at the CEF surfaces. On average, CEFs form a slightly higher number of hydrogen bonds with molecular species in aqueous dilute NaOH:urea than in water (Fig. 4*A*), and this difference arises from the hydrogen bonds between cellulose and urea (Fig. 4*B*). The reduced number of hydrogen bonds between cellulose and water in the NaOH:urea system than water alone is also consistent with the proximal RDF. It has been suggested that urea acts as a hydrogen-bond acceptor from hydroxyl groups on cellulose surface such that the interfibril hydrogen bond is disrupted (26). Our hydrogen-bond (Fig. 4 *C* and *D*) and atom-to-atom RDF analyses (*SI Appendix*, Fig. S6) together show that urea acts as both a hydrogen-bond acceptor and donor with cellulose hydroxyl groups, with the urea carbonyl oxygen accepting hydrogen bonds from the O2 or O6 hydroxyl groups in cellulose, the urea amino nitrogen accepting from the O2 hydroxyl group, and the urea amino group donating to the O6 atom. The existence of the O2:urea hydrogen bond is at the expense of the O2:O6 intrafibril one in the native state and thus indeed destabilizes the fibril. The O6 atom becomes mostly exposed in a liquid medium (42) thereby binding to oxygen in urea. In addition, within the first hydration shell of cellulose, urea has a longer residence time (651 ps) than water in the dilute NaOH:urea (377 ps) and water only (343 ps) in the MD simulations (*SI Appendix*, Fig. S7).

Conclusions

Reducing the high energy cost of fibrillation of cellulose fibers in water is of urgent importance in the production of CNFs for renewable and sustainable biomaterials applications. The use of media in which cellulose fibers are more easily separated than in water alone is a promising avenue in this regard. If a medium is able to reduce the strength of interfiber interactions in pulp pretreatment, it should be easier to shear the fibers apart in subsequent disc-refining fibrillation, thereby reducing energy consumption.

Atomic-detail computer simulation can provide information on interactions permitting the selection of combinations of solvents and polymers for materials processing and hence will reduce a number of expensive wet-lab experiments. Here, we performed all-atom MD simulations of a seven-bundled-CEF model in each

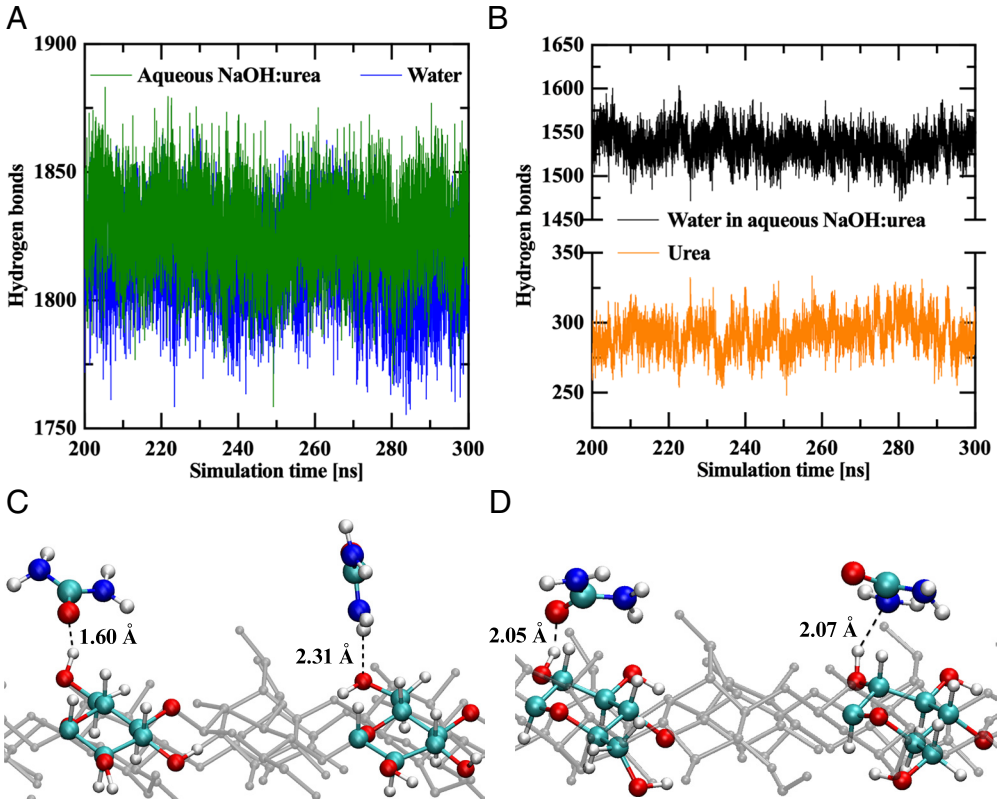

Fig. 4. Time series of numbers of hydrogen bonds between the seven CEFs and (*A*) all species in aqueous NaOH:urea (7:12 wt.%) or sole water; (*B*) water in aqueous NaOH:urea (7:12 wt.%) or urea. The values are averaged over three independent MD runs. (*C*) Two example hydrogen bonds (black dash lines with their distances) simultaneously formed between urea and O6 hydroxyl groups in cellulose. Carbon is cyan, oxygen is red, nitrogen is blue, and hydrogen is white for urea and glucose monomers of interest. Other glucose monomers are transparent gray. (*D*) Similar to (*C*) except for O2 hydroxyl groups in cellulose. The four distances are closely similar to those in the urea-glucose monomer complexes geometrically optimized by density functional theory (DFT) (*SI Appendix*, Fig. S5). Created with BioRender.com.

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of the eight media and in water as a reference. Five of the media reduced the number of inter-CEF atomic contacts as compared with water only, of which, for practical reasons, one (0.007:0.012 wt.% NaOH:urea in water) was able to be selected for pulp pretreatment and pilot-scale fibrillation. Detailed analysis of the MD simulations indicates that urea diffuses into the interfiber space and interacts directly with cellulose fibers by forming hydrogen bonds and increasing interfiber separations. In addition, Zeta potential measurements of the fibrillated CNFs in suspensions show a net mesoscale negative charge which may arise from deprotonation of cellulose hydroxyl groups in the alkaline NaOH environment, leading to electrostatic repulsion between the fibers.

The resulting fiber swelling reduces the energy required for fibrillation in a pilot-scale disc refiner. Experimentally, we observed that using the dilute NaOH:urea in this process saves ~21% energy cost on average. At the preferred fines level of 90% (21), using the NaOH:urea system reduces energy consumption by 777 kW h per metric ton of CNF products, which is close to the average monthly electric energy consumption (899 kW h) by a residential unit in the United States in 2022 (43). Further, NMR, TEM, and tensile testing on fibrillated CNF products all indicate that the NaOH:urea-mediated CNFs have similar crystallinity, morphology, and mechanical strength, respectively, to water-mediated ones.

The present line of exploration could be extended to media with mechanisms of action similar to aqueous NaOH:urea, such as other strong base:small organic molecule with hydrogen-bond acceptor and/or donor capability in an aqueous solution. This may be particularly useful for materials processes that require dispersion of deprotonable polymers [e.g., lignin-based carbon fiber-reinforced composites (44)]. More generally, the present work shows how atomic-detail computer simulation driven by fundamental science can aid experimentalists to design pilot-scale processes for the circular bioeconomy instead of performing a multitude of trial-and-error experiments.

Materials and Methods

Pilot-Scale Fibrillation of Cellulose Fibers in Pretreated Pulp. Bleached softwood kraft (BSK) pulp, which is dried sheets with ~10 wt.% water, was pretreated with either NaOH:urea (0.007:0.012 wt.%) in an aqueous solution or water only as a medium for 12 h to prepare feedstock with ~3 wt.% cellulose fibers. Subsequently, pilot-scale disc-refining fibrillation at six different energy levels (i.e., 0, 2,000, 4,500, 6,000, 7,500, and 8,800 kW h per metric ton of CNFs) was performed in the Process Development Center (PDC) at the University of Maine to produce suspensions with ~3 wt.% CNFs (45). Fines levels of the CNFs in suspensions were measured using a Techpap MorFi Fiber Analyzer (Grenoble, France) based on the Technical Association of the Pulp and Paper Industry (TAPPI) standard T 271 om-02 (46).

Zeta Potential Measurements of Fibrillated CNFs in Suspensions. First, ~20 μL of the suspensions with CNFs fibrillated in either aqueous NaOH:urea (0.007:0.012 wt.%) or water only at 7,500 kW h per metric ton of CNFs were pipetted into a disposable folded capillary cell. Subsequently, the Zeta potential was measured at 25 °C using a Malvern Panalytical Zetasizer Pro (Westborough, MA) with ZS Xplorer Software in automatic mode, where the optimal voltage and number of runs (minimum of 10) are automatically selected after measuring conductivity of the sample.

Crystallinity, Morphology, and Mechanical Strength of CNF Products. NMR. To calculate crystallinity of the CNF products using Eq. [4], the as-prepared suspensions with CNFs fibrillated in either aqueous NaOH:urea (0.007:0.012 wt.%) or water only at 7,500 kW h per metric ton of CNFs were frozen at −40 °C for 12 h, followed by freeze-drying at −40 °C in a Labconco freeze dryer (Kansas City, MO). Solid-state ^{13}C NMR spectroscopy with cross-polarization/magic-angle-spinning (CP/MAS) technique was carried out at 25 °C using a Bruker Avance III 400 MHz (9.4 Tesla) spectrometer (Billerica, MA). The dried CNFs sample was filled in 4 mm MAS zirconia rotors at MAS frequency of 8 kHz. ^{1}H–^{13}C CP contact time was 1.5 ms, recycle delay was 3 s, and number of scans acquired was 3,072.

TEM. To see morphologies of the produced CNFs at nanoscale, the suspensions with CNFs fibrillated in either aqueous NaOH:urea (0.007:0.012 wt.%) or water only at 7,500 kW h per metric ton of CNFs were diluted in CO_2-free water and placed on Cu TEM square grids with 200 mesh to dry. The grids were then positively stained sequentially with 2 wt.% O_sO_4, 3 wt.% lead citrate, and 30 wt.% UranyLess contrast stain solution (Electron Microscopy Sciences, Hatfield, PA) with washing and drying between each staining step. Subsequently, the grids were imaged using a JEOL NEOARM (Peabody, MA) TEM equipment at 120 kV. The sample was imaged at 80 kV, spot size 5, and low beam conditions to minimize changes of the sample induced by the electron beam. Images were acquired with a Blue Scientific Gatan OneView camera (Cambridge, United Kingdom) at 4 k × 4 k resolution using 1-s exposure time.

All-Atom MD Simulations. To model the preparation of feedstock prior to fibrillation in the experiment, which involves pretreatment of pulp (dried cellulose sheets with ~10 wt.% water) in each of the eight examined media or sole water (*SI Appendix*, Table S1) at ambient temperature, two-stage MD simulations were conducted and described in the following two subsections.

Stage 1: Dried Cellulose Fibers to Approximate BSK Pulp. We built an atomistic model of seven bundled cellulose I_β CEFs with a 2-3-2 arrangement (30, 31), each containing 18 glucan chains (3–5) with DP of 20 (~100 Å in length) (32). Each CEF has a side-to-side cross-sectional dimension of ~20 Å (Fig. 1 *A* and *B*). The seven CEFs separated by ~12 Å were solvated in water in a cubic box with an edge size of 170 Å (*SI Appendix*, Fig. S8*A*). We first performed energy minimization, 1-ns NVT equilibration, 1-ns NPT equilibration, and 200-ns NPT production (1 bar and 25 °C) to obtain equilibrated CEFs in water. Subsequently, to simulate cellulose drying in which water evaporates from surface of cellulose fibers (10), we placed the as-equilibrated seven CEFs in the geometric center of a water droplet with 175 Å diameter, containing 80,900 water molecules, and in a cubic box with an edge size of 808 Å (*SI Appendix*, Fig. S8*B*). This box volume V was derived to model saturated vapor of these water molecules as a near ideal gas [Eq. [5]] at temperature T [175 °C (47) to ensure maximum extent of drying], from which pressure P (8.87 bar) was derived using the Antoine equation (48) and compressibility factor Z (0.935) was derived according to a previous study (49).

$$Z = \frac{PV}{nRT}, \qquad [5]$$

where n is the number of moles of water molecules and R is the gas constant. We performed 1-ms NVT simulations (*SI Appendix*, Fig. S8*C*) using long-range electrostatic truncation described and validated in *SI Appendix*, Fig. S9.

Stage 2: Solvation of the Cellulose Fibers from Stage 1 in the Examined Medium to Approximate Pretreatment of BSK Pulp. The as-dried seven CEFs with residual surface water [i.e., water oxygens within 3.6 Å from cellulose (*SI Appendix*, Fig. S2)] were solvated in each of the eight examined media or sole water (*SI Appendix*, Table S1) in a cubic box with an edge size of 170 Å. We performed energy minimization, 1-ns NVT equilibration, 1-ns NPT equilibration, and 300-ns NPT production (1 bar and 25 °C). The last 100 ns were used for analyses once the simulations were found to be approximately equilibrated (*SI Appendix*, Fig. S3). In addition, MD simulations of the CEFs in 31 other media were also performed, and their results are shown and discussed in *SI Appendix*, Table S2. Although these 31 media are not in our final list considering their safety, toxicity, or optimal performance issues at elevated temperatures (23, 50, 51), they could be potentially beneficial to bench-scale fibrillation.

Technical Settings for Computational Work. We used the GROMACS program (52) and the CHARMM36 force fields for cellulose (53, 54) and for small molecules (55, 56) other than water, for which the TIP3P model (57) was used. The force field parameters for γ-valerolactone in Fig. 1*C* and both black and orange underlined medium species in *SI Appendix*, Table S2 were generated by analogy using CHARMM General Force Field (CGenFF) (58, 59). Some of those for the orange underlined medium species required further reparameterization as penalty scores were found to be greater than 50 in CGenFF.

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Reparameterization was conducted using default settings in Force Field Toolkit (ffTK) (60, 61) implemented in VMD (62), where system input files were prepared for Gaussian09 (63) quantum mechanical (QM) calculations including energy minimization and bond or dihedral optimization and their output files were processed. For all MD simulations, three independent runs with different initial velocity distributions were performed to ensure our findings do not depend on the initial configuration. Periodic boundary conditions were used with a time step of 2 fs and LINCS (64) to constrain covalent bonds involving hydrogen atoms. We used a v-rescale thermostat (65) with a 0.2 ps coupling constant with separate temperature groups for the cellulose and medium and Berendsen (66) and Parrinello–Rahman (67) barostats for NPT equilibration and NPT production, respectively, with a 2 ps coupling constant to maintain constant pressure. The instantaneous energies and configurations were saved every 10 ps. The analyses were done using functions built in the GROMACS program (52), and 3 glucose monomers on each end of the glucan chains in the seven-CEF model were not included to approximate μm-long CNFs in experiment (33).

Data, Materials, and Software Availability. All data supporting the findings of this study are included in this article, its Supporting Information, and at https://www.doi.org/10.6084/m9.figshare.26081869 (68).

ACKNOWLEDGMENTS. This manuscript has been authored by UT-Battelle, LLC, under contract DE-AC05-00OR22725 with the United States (US) Department of Energy (DOE). We acknowledge the support from the DOE's Office of Energy Efficiency and Renewable Energy, Advanced Materials and Manufacturing Technologies Office under Corporate Planning System (CPS) agreement 30534 and 35714, and the University of Maine's Hub & Spoke Sustainable Materials & Manufacturing Alliance for Renewable Technologies (SM^{2}ART) Program with the Oak Ridge National Laboratory. This research used resources of the Oak Ridge Leadership Computing Facility (Summit and Frontier supercomputers) and the Center for Nanophase Materials Sciences at the Oak Ridge National Laboratory, which are DOE Office of Science user facilities. An award for the computing time on Frontier was provided by the DOE's Innovative and Novel Computational Impact on Theory and Experiment (INCITE) program. This research used resources of the Compute and Data Environment for Science (CADES) at the Oak Ridge National Laboratory, which is supported by the Office of Science of the US DOE under Contract No. DE-AC05-00OR22725. The US government retains and the publisher, by accepting the article for publication, acknowledges that the US government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for the US government purposes. The DOE will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (https://www.energy.gov/doe-public-access-plan).

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